

Climate Change Report Brochure

Nornickel Group
2024



Vice president's statement



Dear colleagues,

I am pleased to present Nor Nickel's second Climate Change Report, covering the year 2024, in which we detail the results of our efforts to address climate change.

The climate agenda remains a key element of Nor Nickel's development strategy. On the one hand, our assets are concentrated in the Arctic zone, where monitoring climate-related risks is of particular importance. On the other hand, we see strong market potential for our metals, driven by transport electrification, growth in renewables, and the advancement of hydrogen technologies.

We remain committed to reducing our own climate impact. In the reporting year, despite a slight increase in production, we managed to keep gross greenhouse gas emissions

at 7.51¹ mln t – one of the lowest levels among major metals companies. In 2024, renewable electricity accounted for 54% of total onsumption across the Group. Spending on our energy efficiency programme and other climate-related initiatives totalled RUB 4.8 billion.

In addition to the geotechnical monitoring system deployed at the Norilsk site, last year we partnered with Fedorovsky Polar State University to develop a background monitoring system for permafrost soils. Background monitoring will enable us, over time, to better assess trends in permafrost degradation outside urban areas – where a significant ortion of our linear infrastructure is located. In addition, in 2024, we also collaborated with the Institute for Economic Forecasting of the Russian Academy of Sciences to update our proprietary scenarios for global economic and climate development. Based on these scenarios, we revised our orecasts for demand and prices for our metals. Our analysis has shown that under any scenario – whether one of accelerated decarbonisation or of a global retreat from such efforts – our basket of metals ensures the long-term resilience of our financial position.

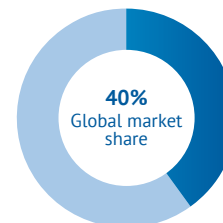
I would like to thank Nor Nickel employees and partners involved in our joint climate-related efforts for their valuable contributions, and I trust that this active cooperation will continue into the future.

With best regards,

Stanislav Seleznev

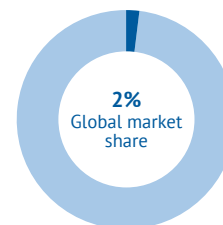
Vice President for Ecology and Industrial Safety
at MMC Norilsk Nickel

About Nor Nickel

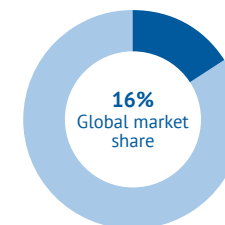


No. 1 palladium
producer

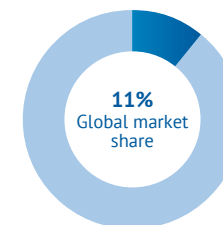
Nor Nickel already supplies four of the nine categories of critical green metals to the global market and is planning to develop a lithium deposit.



No. 13 copper
producer



No. 2 metal nickel
producer



No. 4 platinum
producer

	Nor Nickel's metals					REE	Chromium	Zinc	Aluminium
	Copper	Nickel	Cobalt	PGMs	Lithium				
Solar power	●	●	●	●	●	●	●	●	●
Wind power	●	●	●	●	●	●	●	●	●
Concentrated solar power systems	●	●	●	●	●	●	●	●	●
Geothermal power	●	●	●	●	●	●	●	●	●
Nuclear power	●	●	●	●	●	●	●	●	●
Power transmission lines	●	●	●	●	●	●	●	●	●
Electric vehicles and batteries	●	●	●	●	●	●	●	●	●
Hydrogen energy	●	●	●	●	●	●	●	●	●

Relative importance of metals and minerals for various clean energy technologies

High: ● Medium: ● Low: ●

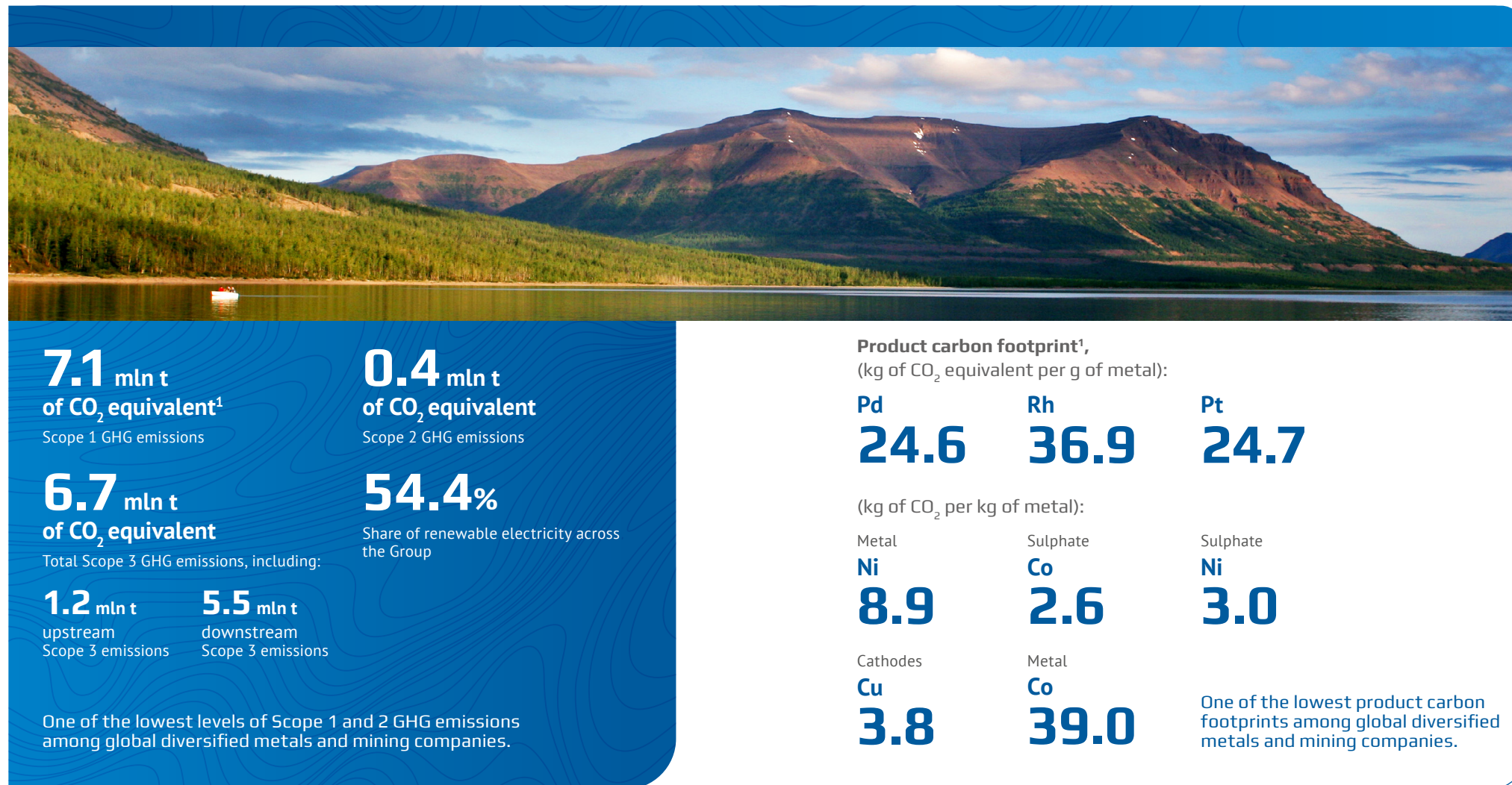
PGMs – platinum group metals. REE – rare earth elements



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¹ Including a GHG emissions provision for the Sulphur Project at Nadezhda Metallurgical Plant and excluding GHG emissions generated from heat and electricity supplies to the public.

Key climate change metrics for 2024



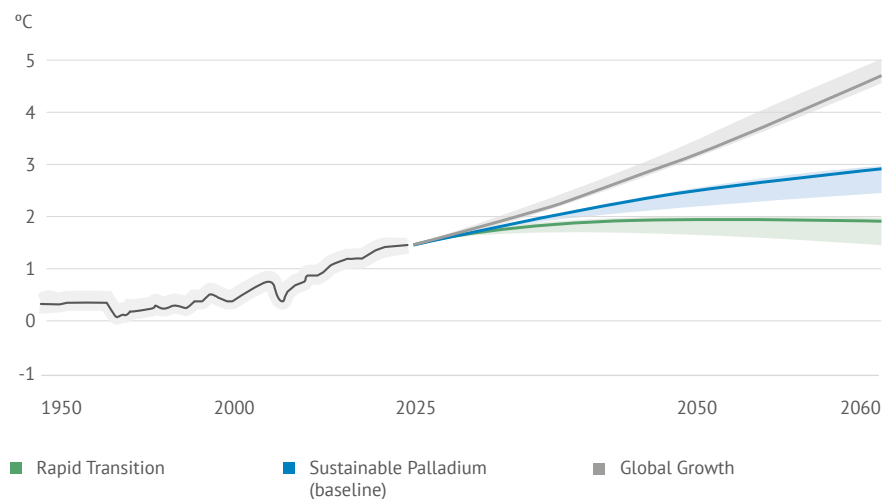
¹ Including a GHG emissions provision for the Sulphur Project at Nadezhda Metallurgical Plant and excluding GHG emissions generated from heat and electricity supplies to the public.

¹ According to ISO 14044, a 100-year global warming potential (GWP) metric was applied. The calculation includes a GHG emissions provision for the Sulphur Project at Nadezhda Metallurgical Plant.

Climate-related risks and opportunities

Transition risks and opportunities:

Nornickel's climate scenarios up to 2060



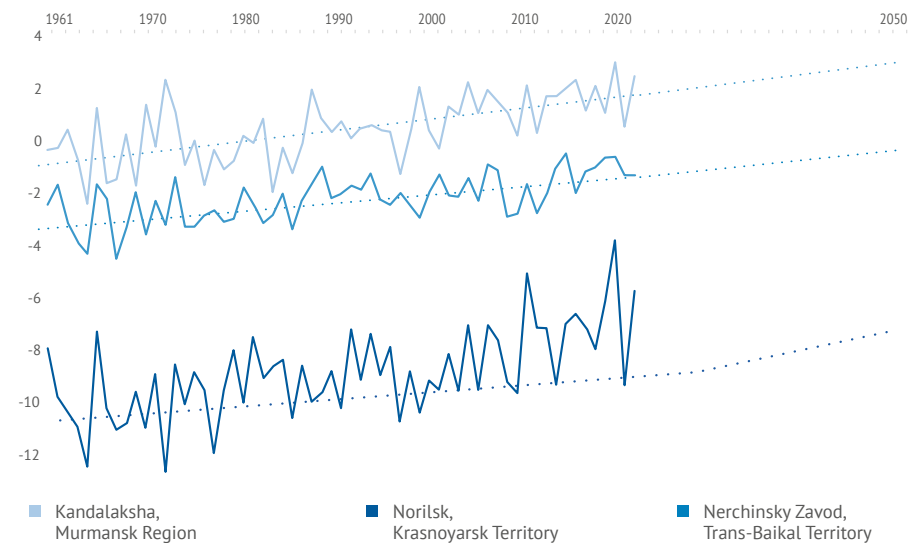
Nornickel collaborated with the Institute for Economic Forecasting of the Russian Academy of Sciences to update scenarios for global economic and climate development up to 2060. The Sustainable Palladium

corporate scenario, which aligns with SSP2-4.5, has been selected as the baseline scenario. Climate stress testing of the financial and economic model indicates Nornickel's long-term business resilience.



Physical risks:

Average air temperature and forecast to 2050 in the Sustainable Palladium scenario, °C



In partnership with the Obukhov Institute of Atmospheric Physics of the Russian Academy of Sciences, Nornickel evaluated

climate change trends across its footprint since the 1960s and developed climate projections up to 2050.

Key risk factors:



permafrost
degradation



precipitation
anomalies



increased
thunderstorm
activity

Decarbonisation

Exploring the absorption capacity of tailings storage facilities

The methodology used to assess the natural absorption of greenhouse gases by gangue in tailings storage facilities has been verified by TÜV AUSTRIA.

Further plans include the study of artificial and enhanced gangue mineralisation in tailings storage facilities with high absorption potential

Removals in 2024

370 kt of CO₂

Construction of solar power plant

Pilot project to supply own energy needs of the rotational camp at Bystrinsky GOK.

500 kWh

Energy efficiency

Effect for Scope 1 and 2 exceeds

>250
kt of CO₂-equivalent



Innovation

Nornickel's Palladium Centre aims to bring over 100 new palladium-containing materials and products to market.

Key R&D:

Green energy

Jet fuel

 Water disinfection

Glass fibre production



Nornickel's Battery Technology Centre focuses on developing nickel-containing cathode active materials – a key component in modern batteries. The first samples of cathode materials for NCM 811+ chemistry have already been developed.

Climate change adaptation

Nornickel's core production site is located in the Arctic, making permafrost degradation the Company's primary climate risk factor. Nornickel is developing a geotechnical monitoring system to mitigate risks to the foundations of buildings and structures built on permafrost.

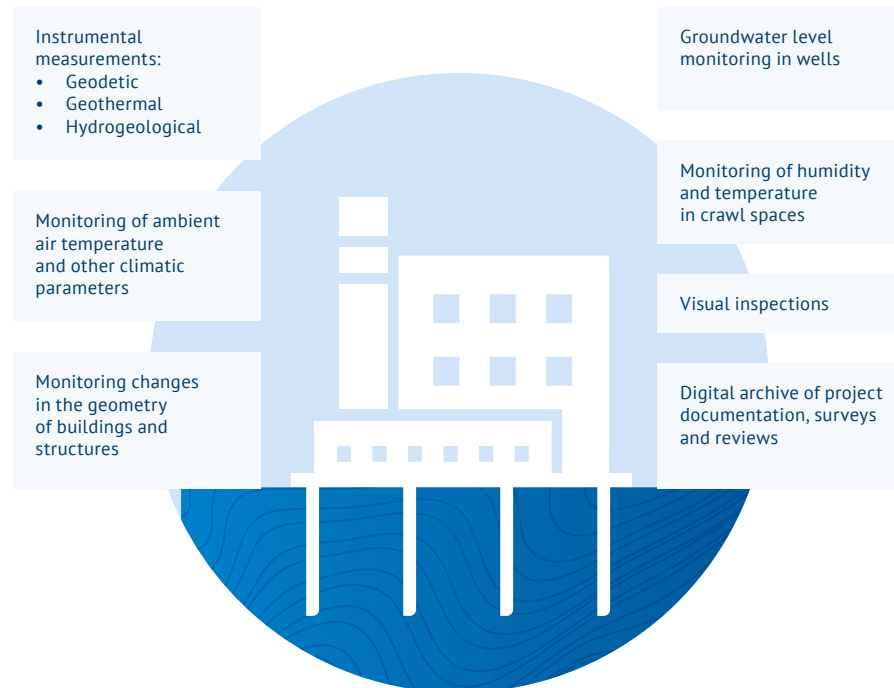
In 2024, to improve the accuracy of risk assessments for linear infrastructure located outside urban areas, Nornickel launched background permafrost monitoring, which complements the geotechnical monitoring system. This initiative will enable more accurate forecasting of permafrost degradation in the region.

Geotechnical monitoring system

Geotechnical monitoring system

Objectives:

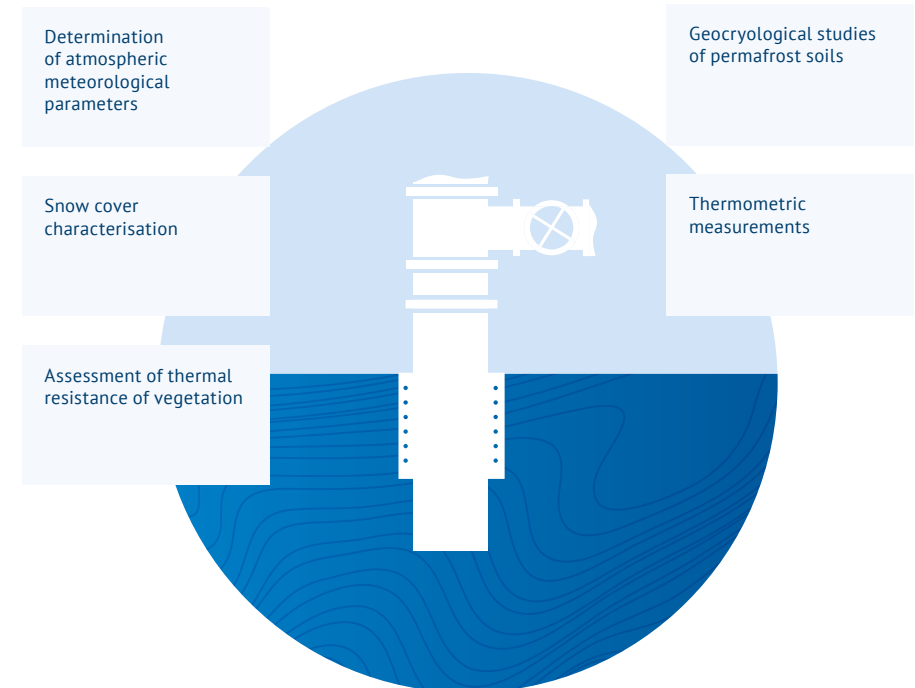
- ensuring the safe operation of buildings and structures (BS)
- assessment of BS condition and forecasting of changes in BS technical condition



Background permafrost monitoring system

Objectives:

- updating permafrost degradation forecasts
- assessing the impact of permafrost degradation on the Company's facilities outside urban areas



Project awards:

- Included in the Best Practices Register, BRICS Solutions Awards 2024
- Winner of "Green Eurasia" international competition in the nomination "Monitoring and Forecasting of Climate Change"
- National Environmental Prize named after V.I. Vernadsky in the nomination "Science for Sustainable development"



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Change Policy](#)